As an aid in visualizing the curvature (warpage) of the star's space, I shall use a drawing called an *embedding diagram* Because embedding diagrams will play a major role in future chapters, I shall introduce the concept carefully, with the help of an analogy.

Imagine a family of human-like creatures who live in a universe with only two spatial dimensions. Their universe is the curved, bowl-like surface depicted in Figure 3.2. They, like their universe, are two-dimensional; they are infinitesimally thin perpendicular to the surface. Moreover, they cannot see out of the surface; they see by means of light rays that move in the surface and never leave it. Thus, these "2D beings," as I shall call them, have no method whatsoever to get any information about anything outside their two-dimensional universe.

These 2D beings can explore the geometry of their two-dimensional universe by making measurements on straight lines, triangles, and circles. Their straight lines are the "geodesics" discussed in Chapter 2 (Figure 2.4 and associated text): the straightest lines that exist in their two-dimensional universe. In the bottom of their universe's "bowl," which we see in Figure 3.2 as a segment of a sphere, their straight lines are segments of great circles like the equator of the Earth or its lines of constant longitude. Outside the lip of the bowl their universe is flat, so their straight lines are what we would recognize as ordinary straight lines.

If the 2D beings examine any pair of parallel straight lines in the outer, flat part of their universe (for example, Lt and L2 of Figure 3.2), then no matter how far the beings follow those lines, they will never see them cross. In this way, the beings discover the flatness of the outer region. On the other hand, if they construct the parallel straight lines L3 and L4 outside the bowl's lip, and then follow those lines into the bowl region, keeping them always as straight as possible (keeping them geodesics), they will see the lines cross at the bottom of the bowl. In this way, they discover that the inner, bowl region of their universe is curved.

The 2D beings can also discover the flatness of the outer region and the curvature of the inner region by measuring circles and triangles (Figure 3.2). In the outer region, the circumferences of all circles are equal to π (3.14159265 . . .) times their diameters. In the inner region, circumferences of circles are less than π times their diameters; for example, the large circle drawn near the bowl's bottom in Figure 3.2 has a circumference equal to 2.5 times its diameter. When the 2D beings construct a triangle whose sides are